

DOCUMENT 1: SOLUTION SYNOPSIS / EXECUTIVE SUMMARY

Project Title: PARIKSHAK-AI (परीक्षक-AI) — Cognitive Copilot & Fairness Architecture for On-Screen Marking
Challenge: Challenge 03: AI-Driven Examination & On-Screen Marking Transformation
Organized by: MPOnline Limited & Department of Higher Education, Government of Madhya Pradesh
Team Name: eMitra (Technical Track)
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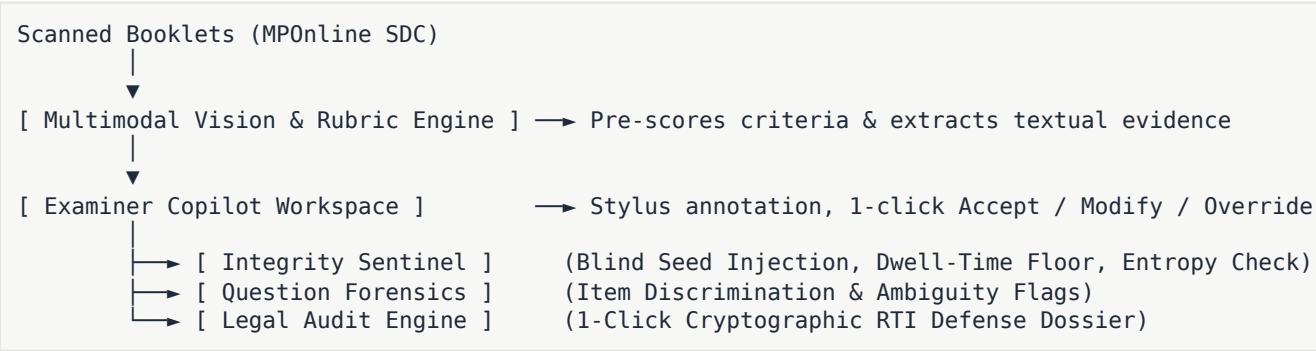
1. Executive Context & The Madhya Pradesh Examination Crisis

Higher education across Madhya Pradesh constitutes one of India's largest state academic ecosystems, enrolling over **28.0 Lakh students** across **2,610+ affiliated colleges** and **~94 universities** (AISHE Data). Each semester, this apparatus processes between **1.4 Crore and 1.7 Crore handwritten answer sheets** (~3.0+ Crore annually). Despite pilot On-Screen Marking (OSM) initiatives, the evaluation system faces recurring administrative crises:

- Catastrophic Quality Failures:** At **Barkatullah University (BU), Bhopal**, discrepancies were suspected in around **30% of results** processed by an outsourced agency: students who sat the exam were marked absent, given zero marks or shown subjects they never took, and nearly **10,000 complaints** followed. At **Devi Ahilya Vishwavidyalaya (DAVV), Indore**, only about **43% of ~7,200 B.Ed candidates** passed a first-semester exam, and allegations of faulty evaluation forced an expert re-evaluation of answer sheets.
- The "3-Minute Valuation" Reality:** Between 15,000 and 25,000 empanelled faculty examiners are paid a meager **₹12 to ₹20 per answer booklet**. Mandated to evaluate 40–50 scripts of 36 pages each within a 4-hour window, examiners spend barely **180–300 seconds per booklet**, resulting in superficial "glance-checking", severe cognitive fatigue, and arbitrary marking.
- Result Paralysis & The Revaluation Toll:** Delays range from **3 to 14 months**, forcing final-year students to forfeit campus placement offers, UPSC/MPPSC recruitment cut-offs, and postgraduate admissions. Concurrently, universities collect an estimated **₹30 to ₹45 Crore annually** in non-refundable revaluation fees from distressed students protesting flawed grading.

2. The Solution: PARIKSHAK-AI (परीक्षक-AI)

PARIKSHAK-AI is an enterprise-grade **Human-in-the-Loop (HITL) Cognitive Copilot** specifically engineered to augment MPOnline's established examination ecosystem. Rather than attempting to replace human faculty—which violates statutory university ordinances and UGC norms—PARIKSHAK-AI serves as a transparent cognitive assistant that pre-reads, rubrics-aligns, and calibrates every script while preserving final decision authority with human examiners.



3. Five Transformative Innovations

1. Native Indic Multimodal Ingestion (Devanagari + Hinglish):

An estimated 60%+ of MP state university students (BA, B.Com, B.Sc, B.Ed, LLB streams) write in Hindi. Traditional OCR pipelines fail because segmentation errors cascade into grading errors. PARIKSHAK-AI uses end-to-end multimodal vision to evaluate handwritten Hindi, English, and technical drawings directly against analytic rubrics without brittle OCR dependencies.

2. Blind Seed-Script Calibration (Cambridge / IB Quality Standard):

To eliminate inter-rater variance, Chief Examiners pre-score 5 "Anchor Scripts" across performance bands. These are invisibly interspersed into examiner queues. If an examiner's mark drifts more than 15% of the maximum marks from the Chief Examiner's mark, the CoE is alerted and the examiner's next scripts are silently routed to the Head Examiner before erroneous marks propagate.

3. Cognitive Velocity & Dwell-Time Sentinel:

Sets a reading floor for each answer from its length, equations and diagrams ($T_{min} = (N_{words}/200 + 0.5 N_{eq} + 0.75 N_{diag}) \times 60 + 10s$). A rushed submission first gets a soft prompt, then the examiner must check at least one rubric criterion; repeated speed-checking is silently routed for a second reading. The examiner is never locked out.

4. Live Question Paper Psychometric Forensics:

Computes item-analysis statistics (difficulty and point-biserial discrimination) across the student cohort as marks arrive. If Question 3(b) produces an 85% failure rate among top-quartile students, it is immediately flagged as ambiguous or out-of-syllabus, enabling the Controller of Examinations (CoE) to issue standard moderation before results publish.

5. 1-Click RTI Cryptographic Justification Dossier:

Following the June 2026 MP High Court judgment that upheld digital evaluation and recommended stylus-based marking, PARIKSHAK-AI generates an instant tamper-evident PDF showing the student's answer with the examiner's annotations, the model key, the evidence behind every mark, examiner timestamps, and a signed SHA-256 Merkle root that supports a certificate under Section 63 of the Bharatiya Sakshya Adhiniyam 2023.

4. Feasibility, Cloud Sovereignty & Economics

- **Non-Invasive Integration:** Seamlessly integrates with MPOnline's existing ASP.NET / Oracle examination portal via secure REST APIs and pre-signed storage URLs without altering core ERP modules.
- **Affordable Unit Economics:** AI inference costs an estimated ₹0.5–₹3 per booklet at current Gemini Flash-Lite/Flash list prices, a fraction of current revaluation overhead. Our proof of concept logs the real token count and cost of every call.
- **Compliance by Design:** Aligned with NEP 2020 (Sections 4.34–4.37) and designed for the DPDP Act 2023 and MeghRaj / MP State Data Centre requirements; in production, inference runs on a model hosted in the State Data Centre or an India-region cloud.
- **Working Proof of Concept:** The examiner workspace, AI pre-read of Hindi/English answers, reading-time check, blind seed scripts and signed audit dossier already run (repository: <https://github.com/nishchaydev/Mphack>).